

iii) Hill Cipher

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Multiletter Cipher

Encrypt a group of letters based on the key value.

Diagram - Digraph, Trigraph, Polygraph

Digraph - Two letters at a time

Trigraph - Three " " " " "

Polygraph - Multiple " " " " "

Encryption :-

Method 1 - PK method

Method 2 - KP method (Tutorial)

Method - 1

$$C = E(P, K) \Rightarrow P \times K \text{ mod } 26$$

Section size = 3

$$(C_1, C_2, C_3) = (P_1, P_2, P_3) \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix} \text{ mod } 26$$

PT: Pay more money
pay mor emo ney.

P a y m o r e n

15 0 24 12 14 17 4 13

a = 0 d = 3 g = 6 j = 9 m = 12 P = 15 S = 18

b = 1 e = 4 h = 7 k = 10 n = 13 q = 16 t = 19

c = 2 f = 5 i = 8 l = 11 o = 14 r = 17 u = 20

v = 21 x = 23 z = 25

w = 22 y = 24

K = $\begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$

$(C_1, C_2, C_3) = (P_1, P_2, B_2)$ $\begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix} \text{ mod } 26$

$C_1 = (P, a, y)$ $\begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix} \text{ mod } 26$
 $[15, 0, 24)$

$$15 \times 17 + 0 \times 21 + 24 \times 2 = 255 + 48 = 303 \text{ mod } 26 = 17$$

$$15 \times 17 + 0 \times 18 + 24 \times 2 = 255 + 48 = 303 \text{ mod } 26 = 17$$

$$15 \times 5 + 0 \times 21 + 24 \times 19 = 75 + 456 = 531 \text{ mod } 26 = 11$$

Pay = RRL

$$C_2 = (m, o, r)$$

17	21	5
21	18	21
2	2	19

$$12 \times 17 + 14 \times 21 + 17 \times 2 = 204 + 294 + 34 = 532 \text{ mod } 26 = 1$$

$$12 \times 17 + 12 \times 18 + 17 \times 2 = 204 + 216 + 34 = 454 \text{ mod } 26 = 22$$

$$12 \times 5 + 14 \times 21 + 17 \times 19 = 60 + 294 + 323 = 677 \text{ mod } 26 = 1$$

mor = MWP

$$C_3 = (e, m, o)$$

17	17	5
21	18	21
2	2	19

$$4 \times 17 + 12 \times 21 + 14 \times 2 = 68 + 252 + 28 = 348 \text{ mod } 26 = 10$$

$$4 \times 17 + 12 \times 18 + 14 \times 2 = 68 + 216 + 28 = 312 \text{ mod } 26 = 0$$

$$4 \times 5 + 12 \times 21 + 14 \times 19 = 20 + 252 + 266 = 538 \text{ mod } 26 = 18$$

emo

$$emo = KAS$$

$$C_4 (x, y) \\ (13, 4, 24)$$

$$\begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$$

$$13 \times 17 + 4 \times 21 + 24 \times 2 = 221 + 84 + 48 = 353 \text{ mod } 26 = 15$$

$$13 \times 17 + 4 \times 18 + 24 \times 2 = 221 + 72 + 48 = 341 \text{ mod } 26 = 13$$

$$13 \times 5 + 4 \times 21 + 24 \times 19 = 65 + 84 + 456 = 605 \text{ mod } 26 = 7$$

$$ney = PDH$$

pay

mor

emo

ney

RRL

MWB

KAS

PDH

4-6-26 Method 2

$$\begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix}$$

mod 26

iv) Polyalphabetic Ciphers

PT: we are discovered save yourself

Key: deceptive

PT length = 21

Key length = 9

So repeat the key 3 times

deceptive deceptive deceptive

wearedisc oversavey ourself

The intersect^{value} of key and PT is CT.

b) Transposition Techniques

Change Position

i) Rail Fence Transposition

Write PT as series of letter place in as diagonal:

PT: Meet Me after the toga party

CT: m e m a t o h t g p r y
e t e f e t e o a a t

CT: mematohtgpryeteeteoaaat

ii) Row Transposition Ciphers

Write PT in a rectangle, row, column

PT: a t t a c k
p o s t p o
n e d u n t
i l t w o a
M V W X Y Z

key: 3 2 4 6 1 5

CT: t s d t w t o e l v a t u w x k o t a z
a p n i m c p n o y

iii) Rotor Machine

3 cylinders each consist of 26 letters

Fast, medium, slow

If first cylinder completes a circle the second cylinder moves 1 step

5-2-26

Module - 2

Conventional Symmetric key Encryption

Symmetric key Encryption

- Same key
- Fast and efficient
- Bulk data encryption
- Major challenge: key distribute
- Two main category: Block Cipher
Stream Cipher

Block

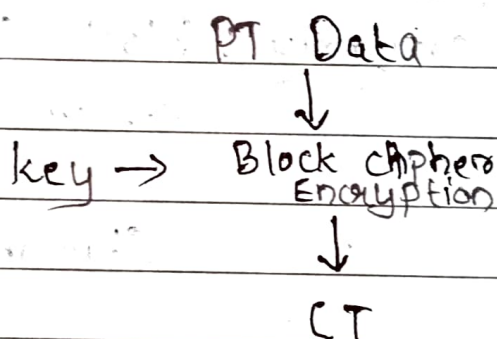
Stream

- | | |
|--|--|
| <ul style="list-style-type: none">• Fixed size blocks• Block size > 64 bit• Same key used for each block• Eg:<ul style="list-style-type: none">AES (128 bit)DES (64 bit) | <ul style="list-style-type: none">• Bit by bit or byte by bytes• Faster than block cipher• Key stream generations.• Use Cases: SSL/TLS connection
Bluetooth
cellular and 4G
Eg: RC4 |
|--|--|

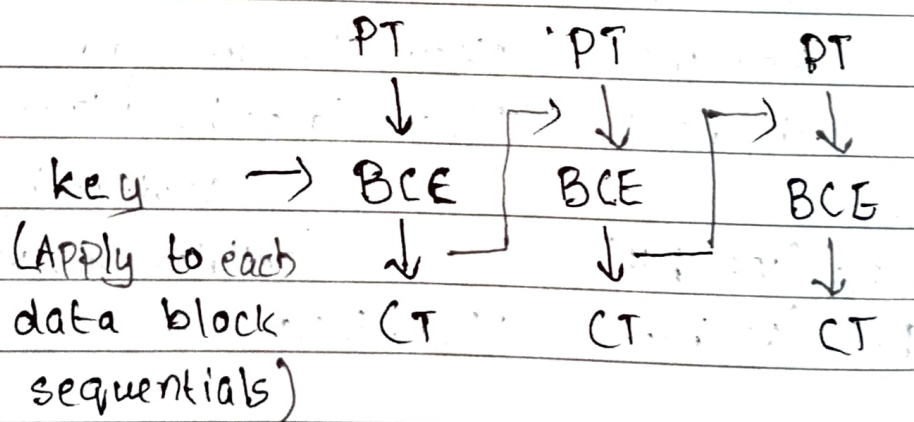
Block Cipher

- Predetermined size of blocks
- Encrypt all the data in a block as a single unit.
- PT → binary → divided into block.
- 0 can be added for excess bits.

Basic Block Cipher Works:-



Basic block chain operation work:-



Ex:

DES , RCS

AES , Triple DES

Blowfish , IDEA Two fish

Advantage : Block cipher are not breakable.

Stream Cipher

→ Encrypt data in long, pseudorandom streams.

→ Process one bit of data at a time.

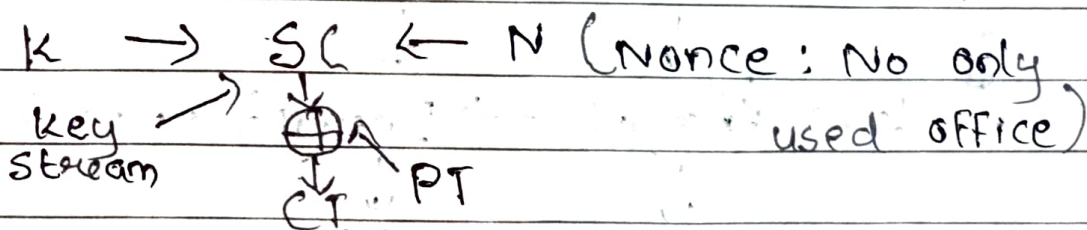
→ Two Types :

Synchronous - (KAK key / autokey)
Automatically key generated

Asynchronous - (CTAT)

It depends on previous PT & CT

6-2-26 Basic Stream Cipher Works :-



Block cipher modes of operation:-

How to securely encrypt and decrypt large amount of data using a block cipher.

Electronic Code Book (ECB)

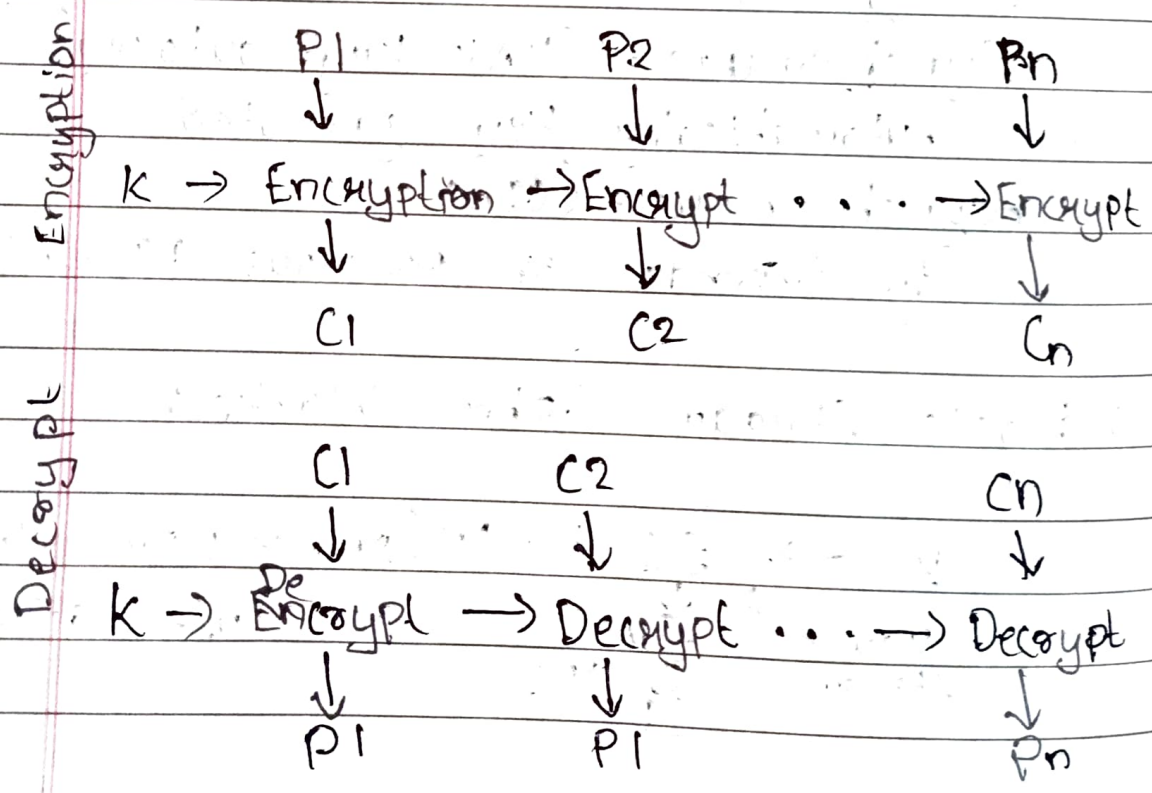
Cipher block chaining (CBC)

Cipher feedback Mode (CFB)

Output Feedback Mode (OFB)

Counter Mode (CM) (CTR)

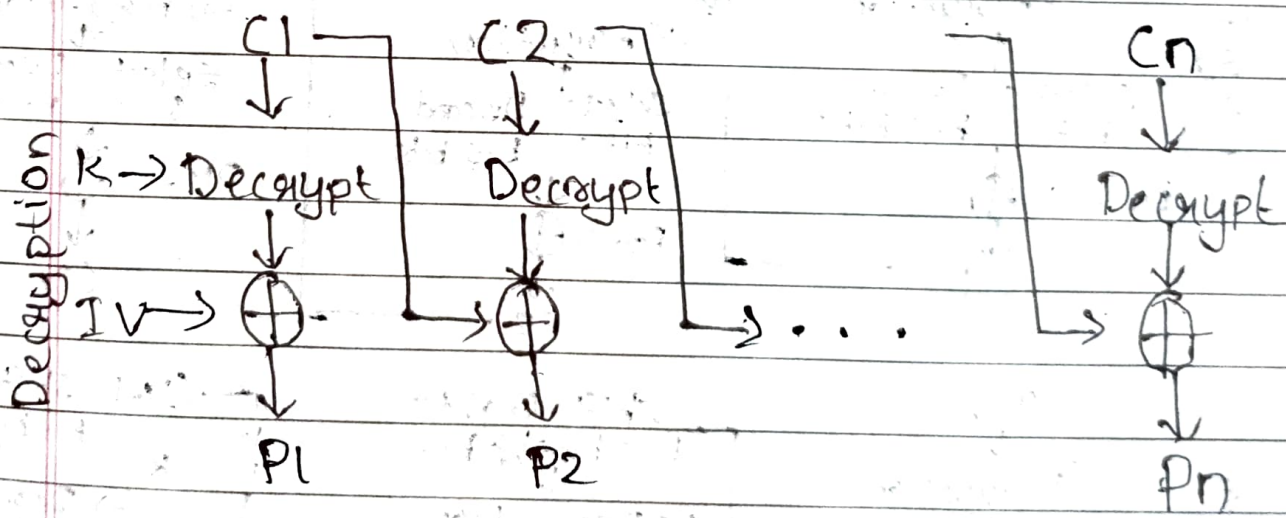
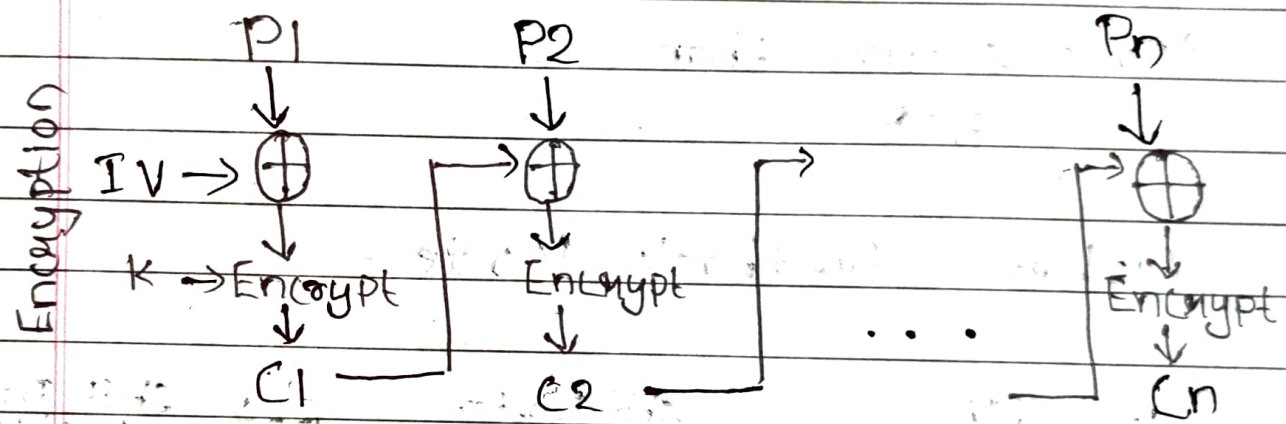
Electronic Code Book -



Advantage
Parallel encryption
Faster & Simple way

Disadvantage
Cryptanalysis occurs
Same PT block

Cipher Block Chaining -

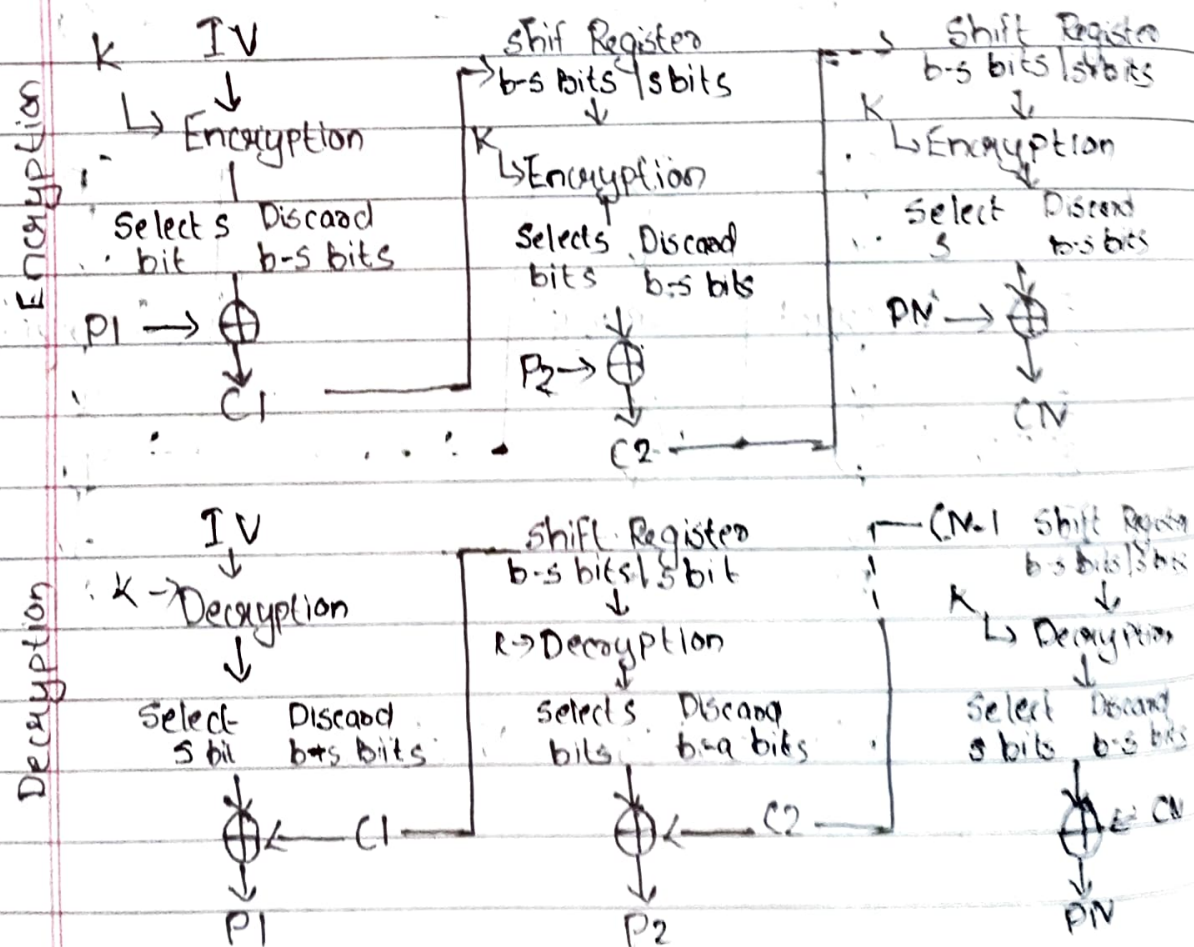


IV → Initialization Vector

Advantages
 Good authentication
 mechanism
 CBC Work well
 Better resistive
 nature
 More secure than
 ECB

Disadvantage
 Previous CI block
 Parallel processing
 is different for
 energy.

Cipher Feedback Mode -



Advantage

Crypt analysis is difficult

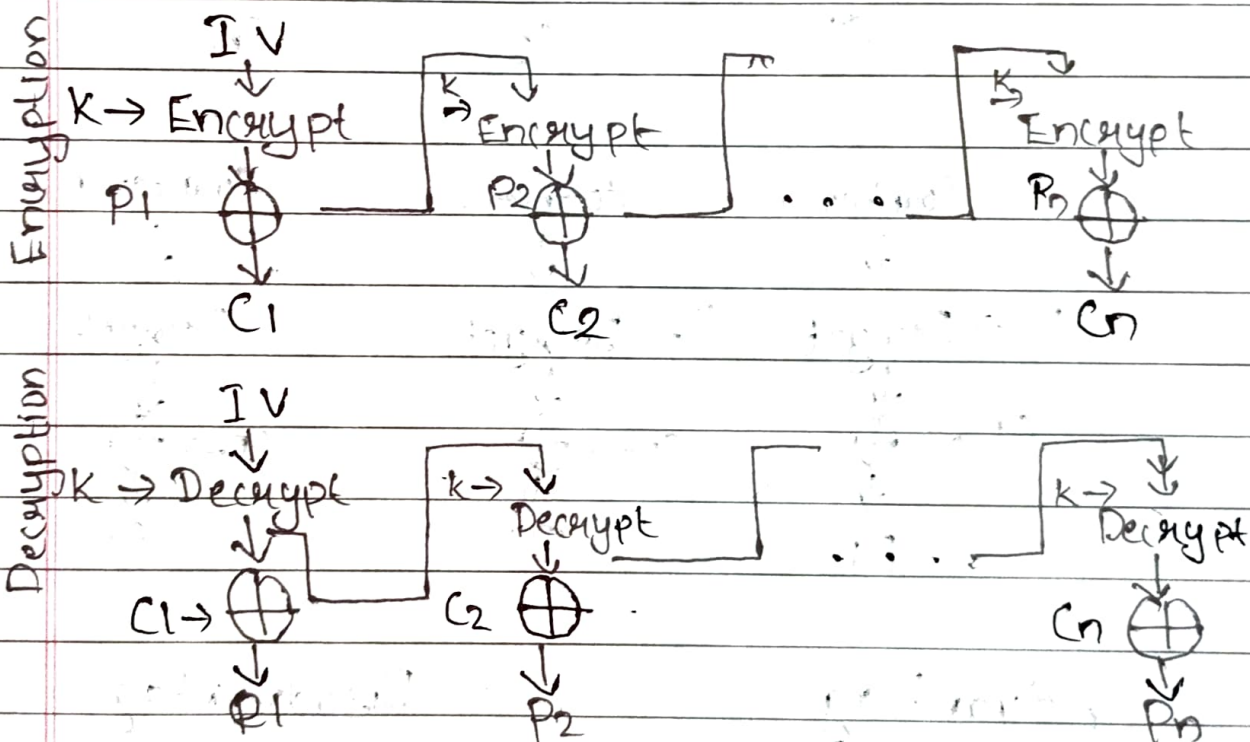
It can handle any size of data

Disadvantage

Parallel processing is not possible

Complex, more chance of error.

Output FeedBack Mode



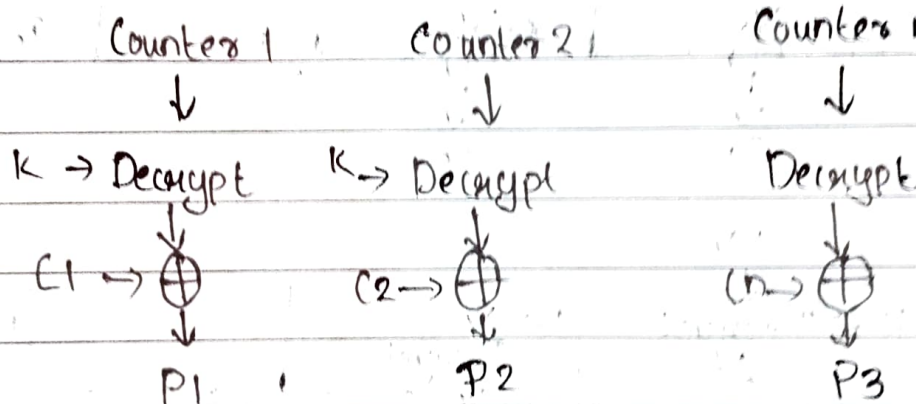
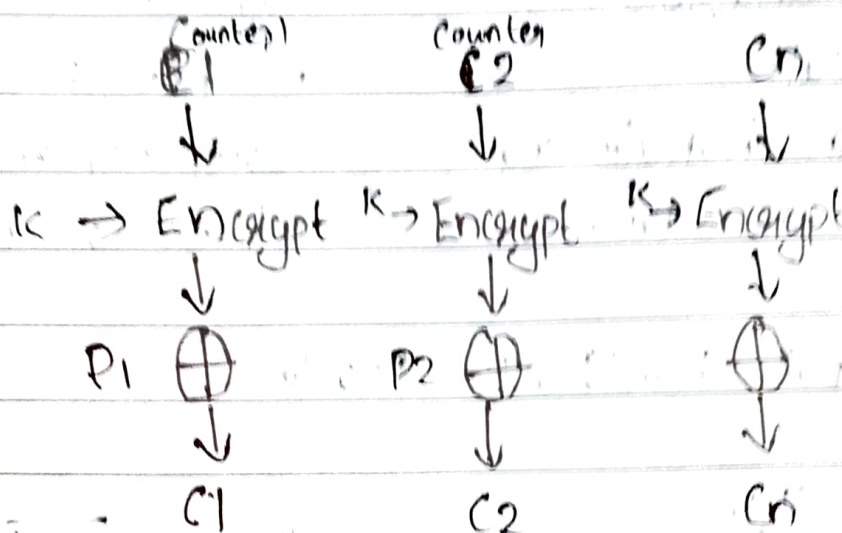
Advantage

Less error propagation

Disadvantage

OFB is simple to attack than CFB
key stream is used

Counter Mode -



Advantage

Disadvantage

Same PT can map
to different CT
Parallel execution
of encryption

The recovery of PT
is inaccurate when
synchronization is
less

9-02-26 Feistel Cipher Structure

Block & key size is 64 bit

S-P network (Substitution - Permutation)

Two primitive operations : Substitution
Permutation

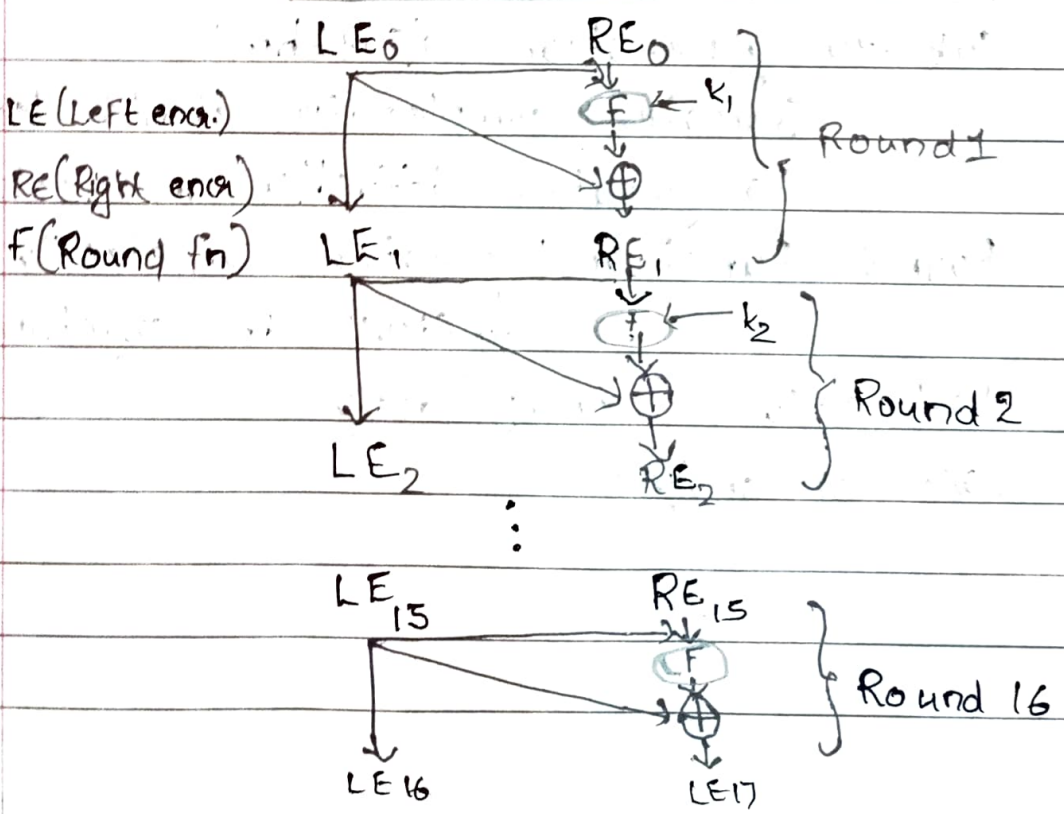
Provide confusion & Diffusion of message & key.

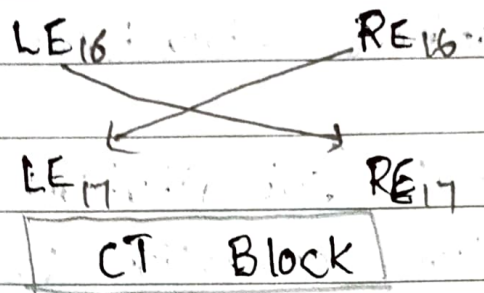
Confusion : make relationship b/w CT and key as complex as possible

Diffusion : dissipates statistical structure of plaintext over bulk of CT change
1 bit it will affect

Feistel structure (Encryption) :

PT : Block





16 round of substitution

1 round of permutation

Round 17 has same structure

Feistel Cipher Design Element:-

Block size : Used larger block size - ^{security} greater

Key size : larger key size - greater security

No. of rounds : multiple round - " "

Subkey generation algorithm:

Greater complexity should lead to greater difficulty in cryptanalysis.

Round function : greater complexity

difficult in cryptanalysis.

Fast software encry/decry

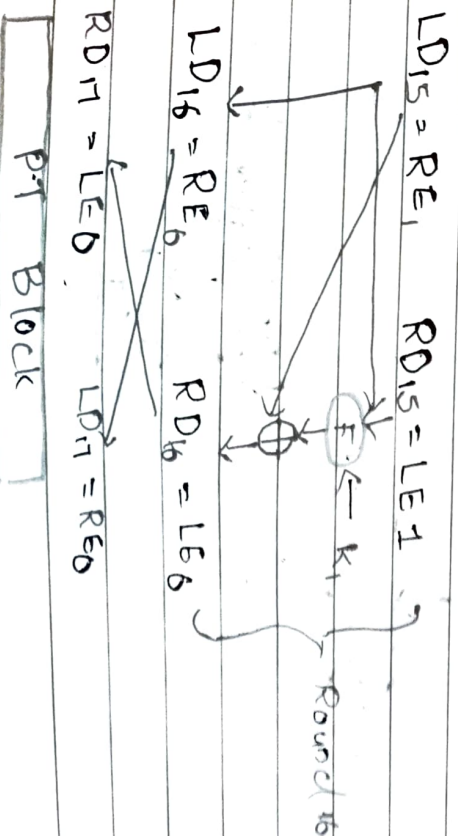
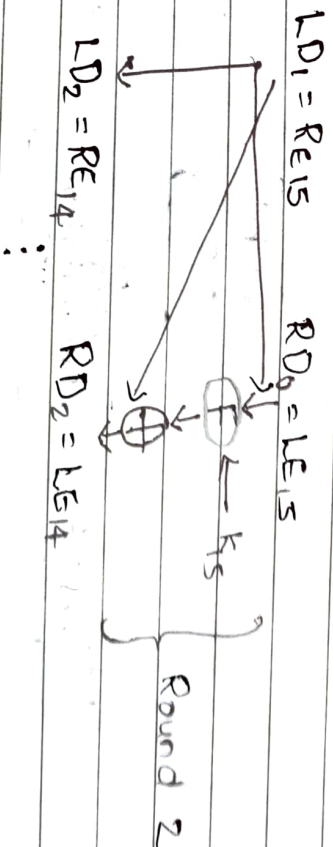
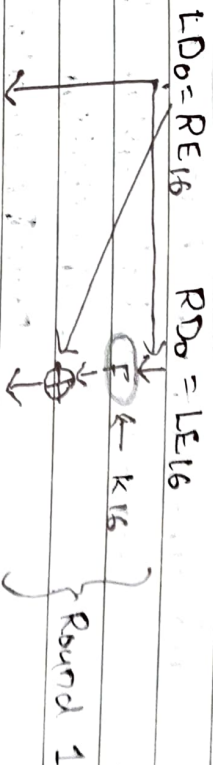
Ease of analysis

Feistel structure (Decryption) :-

I/P 1-CT

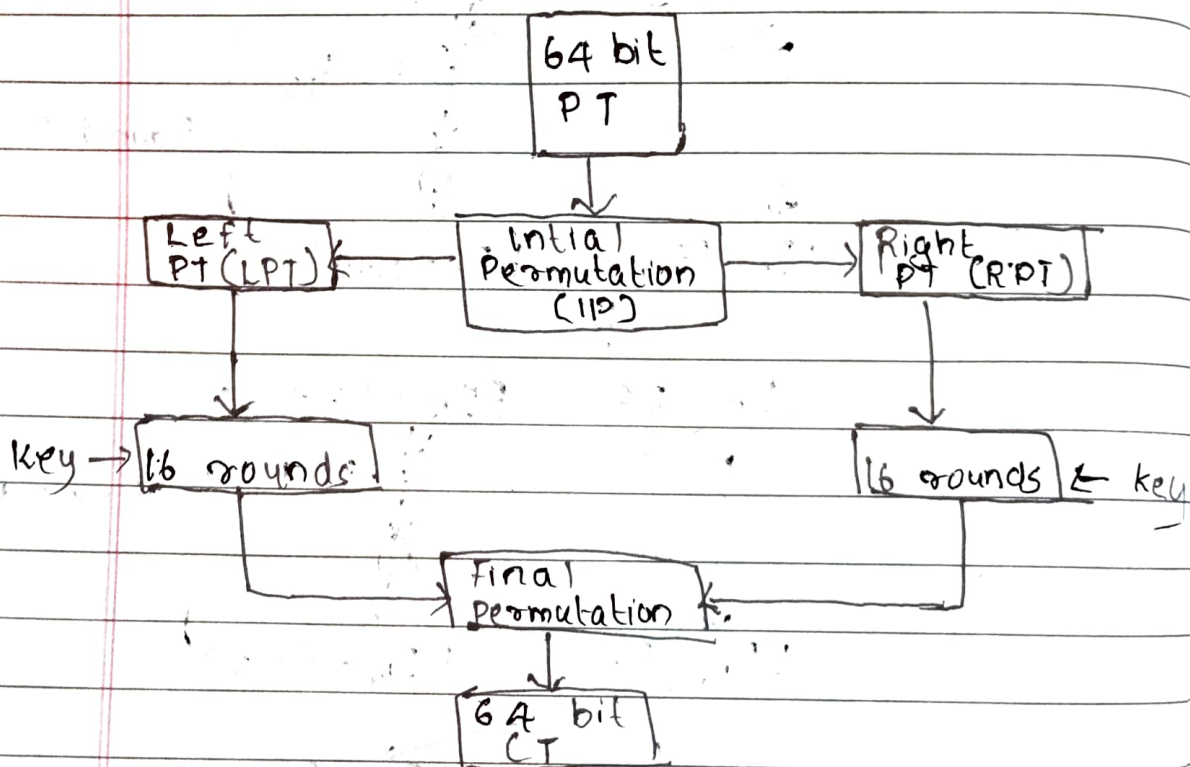
subkeys in reverse order
 k_n first round k_{n-1} in second round k_1 in last round

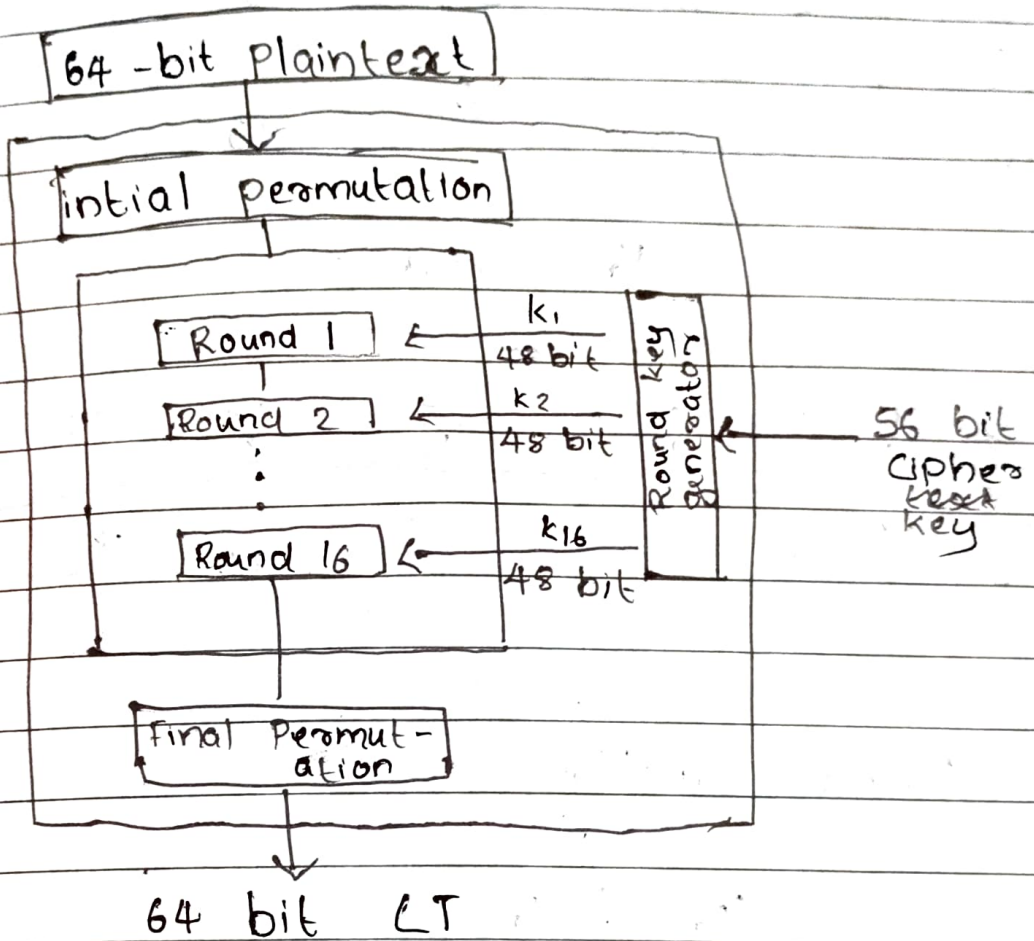
CT Block



DES Data Encryption Standard

- Symmetric key block cipher
- National Institute of Standards and Technology (NIST)
- Block Size 64 bit
- Key size 56 bit (Remove 8th bit)
- Implementation based on Feistel cipher
- Two fundamental mechanisms:
 - Substitution (confusion)
 - Permutation (diffusion)

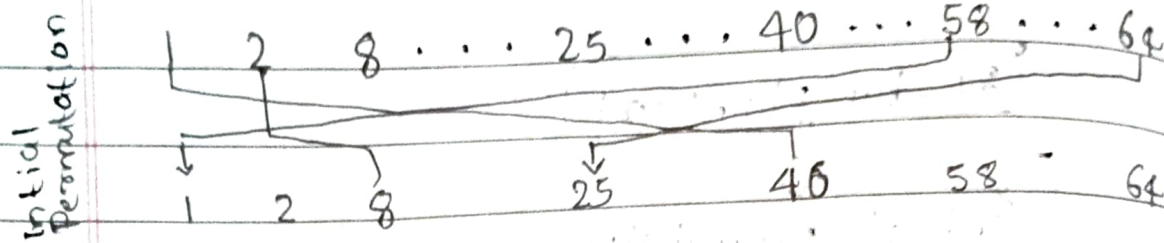




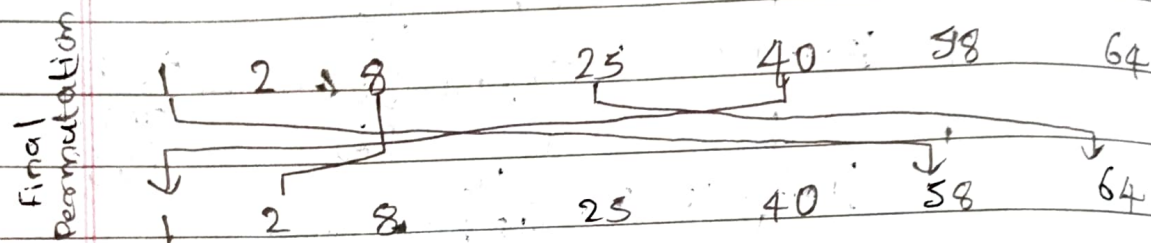
13-02-26 DES has three requirements :

- Round Function
- Key Schedule
- Initial and Final permutation.

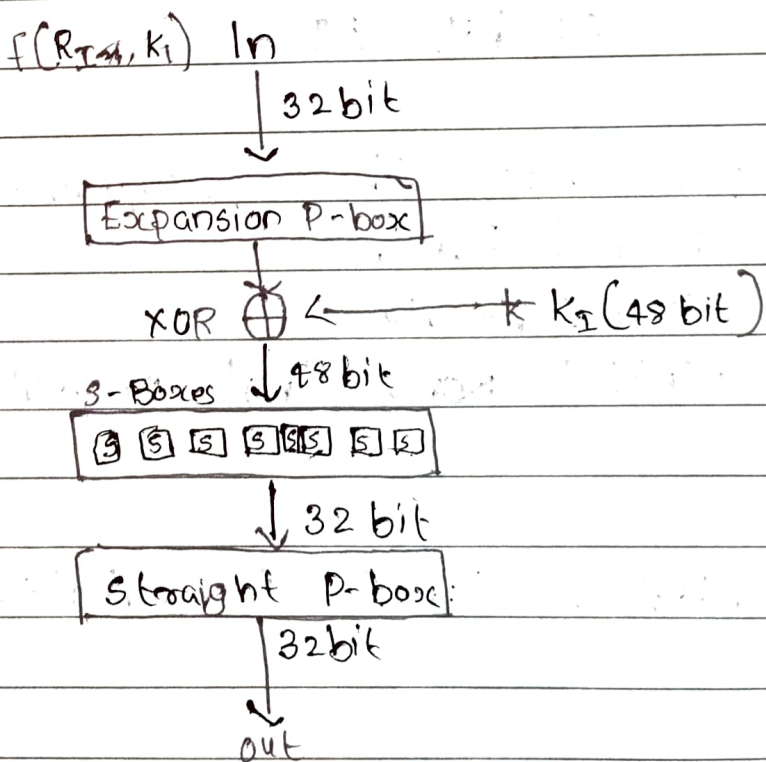
Initial & Final Permutation.



16 rounds



Round Function



Expansion Permutation Box :-

32 bit $\rightarrow$ 48 bit

XOR (Whitener) :-

plaintext XOR with key
48 bit 48 bit

Substitution Box :-

8 boxes it has 8 input

8 inputs reduces to 4 bit it makes

$4 \times 8 = 32$ bit.

